

SPEEDY TRAVEL

MAGLEV TRAINS

Maglev (magnetic levitation) trains have no engine or wheels. Instead, they use powerful magnets to float in the air above the track, while more magnets on the track pull them forwards. The lack of friction between the train and the track allows the train to speed up quickly. The train and tracks also undergo less wear and tear, so they need little maintenance compared to conventional trains.

Floating in air

Most maglev trains use special types of magnets called electromagnets to levitate themselves. These are located at the side of and underneath the train and repel other electromagnets on the guiding rail. To propel the train, an electric current is passed through coils into the guiding rail. This creates a changing magnetic field so that the magnets on the train are constantly being repelled and attracted, moving it forward.

The nose of the train is curved to produce an aerodynamic shape.

The guiding rail is what the train travels over. The train body wraps around it.

The Shanghai Maglev train is the **fastest train** in the world, with a top speed of **267 mph (430 km/h)**.

